6. Given an arbitrarily long character string, you are asked to find out the most frequently occurring character and the least frequently occurring character. Write a C function to solve this problem, given a character string of size N. Now you are told that instead of being a fixed size character string, you are given a stream of characters that are coming dynamically as input. At any point in time, you should be able to print the most frequent and least frequent characters in the stream (note that if two or more characters have the same highest frequency, you can choose to print any of them as the most frequent characters). What is the time and space complexity of your algorithm?
7. You are given two sentences S1 and S2, each containing N and M words, respectively. You are now asked to convert S1 into S2 by either deleting a word or adding a new word. Modifying word1 to word2 can be considered as word deletion followed by word addition. Each insert operation has a cost of +2 and each delete operation has a cost of +1. Given two arbitrary sentences S1 and S2, the edit distance of (S1, S2) is the minimum total cost of operations needed to convert S1 into S2. You are asked to write a function to compute the edit distance. What is the time and space complexity of your function?
8. You have written a program which takes N seconds on a single processor system. Now you have been told to reduce the execution time of the application. So you decided to parallelise the application and run it on an 8-processor system. However, you found that only 70 per cent of the application can be parallelised. What is the speed-up you can expect when you run the modified code on the 8-processor system?
9. You have been asked to write code to reverse a singly linked list. You wrote a non-recursive version, which can do this task. However, the interviewer contends that your solution is not good and a recursive solution would be better than a non-recursive solution, in terms of time and space complexity. Do you agree with the interviewer? If yes, explain why? If not, can you explain the time and space complexity of non-recursive and recursive versions for reversing a linked list?
10. A couple of months back, a popular news item doing the rounds in the tech social media space was about a candidate getting dropped by a Google technical recruiter, based on some questions for which he was expected to give a stereotypical answer. The post makes for hilarious reading at: *http://www.gwan.com/blog/20160405.html*. However, some of the questions in the post are pretty relevant in understanding the candidate's conceptual understanding (though it requires that the recruiters should also know computer science and not just read from a script sheet that they have been given). One of the questions is about the efficiency of sorting algorithms. When would you prefer Quicksort to Mergesort?
11. You are given a document corpus consisting of more than a 100,000 documents, with each document containing at least 1000 words. Given a document D, you are asked to find all documents that are similar to D. How would you go about doing this?
12. In machine learning, there are different classification methods such as Naïve Bayes, Support Vector Machines, Random Forests, etc. Given a particular data set and a binary classification problem, how would you decide which classifier to try on the data set first? Does your selection of the classifier depend on the characteristics of the data set? If your answer is 'Yes', explain why. If not, justify why it does not depend on the data set.
13. What is meant by multi-task learning? Can you explain the benefits of multi-task learning?
14. You are given two singly-linked lists A and B, containing integers. You are asked to find the union and intersection of the two linked lists and represent the results in (a) linked list format, and (b) in a set format. What is the time and space complexity of your solution?
15. You are given an array A of N integers and a number K. Find a pair of integers in the array A such that the sum of the pair of integers is closest to K. What is the time complexity of your solution if (a) the given array A is unsorted, and (b) the given array A is sorted?
16. In operating systems, what is the difference between a deadlock and livelock? How would you distinguish one from the other?
17. In UNIX, can you explain the difference between fork and exec system calls?
18. What is the halting problem? Can you sketch a quick proof for the halting problem? Why is the halting problem so important in computer science?
19. Can you explain, conceptually, the difference between a mutex, semaphore and a barrier?
20. What is an atomicity violation? Can you give a small coding example of it?

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming! END

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